Locally Convex Topological Vector Spaces

Definitions and Examples

Definition: A topological vector space (TVS) is a vector space X with a topology $\mathcal{T}$ such that, with respect to this topology, the maps

(i) $X \times X \to X$, $(x, y) \mapsto x + y$

(ii) $\mathbb{F} \times X \to X$, $(\alpha, x) \mapsto \alpha x$

are continuous with respect to $\mathcal{T}$.

Example: If $\mathcal{P}$ is a family of seminorms on X, and $\mathcal{T}$ is the topology on X with subbasic sets $U(x_0, p, \varepsilon) = \{x \in X : p(x - x_0) < \varepsilon\}$, then $(X, \mathcal{T})$ is a topological vector space.

Definition: A locally convex topological vector space (LCS) is a TVS equipped with a family of seminorms $\mathcal{P}$ that satisfy

$$\bigcap_{p \in \mathcal{P}} \{x : p(x) = 0\} = \{0\}.$$

That is, the family $\mathcal{P}$ separates the points of X.

Proposition: Let X be a TVS, and let p be a seminorm on X. The following are equivalent:

(a) p is continuous;

(b) $\{x \in X : p(x) < 1\}$ is open;

(c) $0 \in \{x \in X : p(x) < 1\}^{\circ}$;

(d) $0 \in \{x \in X : p(x) \leq 1\}^{\circ}$;

(e) p is continuous at 0;

(f) there is a continuous seminorm q on X such that $p \leq q$.

Proof. We see immediately that (a) implies (b) since the set is equal to the inverse image of the open subset $[0, 1) \subseteq [0, \infty)$. Similarly, if $p^{-1}([0, 1))$ is open, then it is equal to its own interior, and since $0 \in p^{-1}([0, 1))$, it follows that there is a subbasic open neighborhood of 0 contained in $p^{-1}([0, 1))$, giving (c). Similarly, we obtain (d) by the fact that $0 \in p^{-1}([0, 1))^{\circ} \subseteq p^{-1}([0, 1])^{\circ}$.

Next, we see that (d) implies that for every $\varepsilon > 0$, $0 \in p^{-1}([0, \varepsilon])^{\circ}$, so if $(x_i)_i$ is a net in X converging to 0 and $\varepsilon > 0$, there is some i_0 such that for all $i \geq i_0$, $x_i \in \{x : p(x) \leq \varepsilon\}$. That is, $p(x_i) \leq \varepsilon$ for all $i \geq i_0$, whence p is continuous at 0. This gives (e).

If p is continuous at 0, then if $x_i \to x$ is a convergent net in X, the reverse triangle inequality gives $|p(x) - p(x_i)| \leq p(x_i - x)$, whence $p(x_i) \to p(x)$, so p is continuous. This gives that (e) implies (a).

Taking $q = p$ yields (a) implies (f). We will show that (f) implies (e).If $x_i \to 0$ in X, then $q(x_i) \to 0$, but since $0 \leq p(x_i) \leq q(x_i)$, it follows that $p(x_i) \to 0$. □

Example:

(a) Let X be a completely regular topological space, and let $C(X)$ be the set of all continuous functions from X to $\mathbb{F}$. If K is compact in X, we may define a seminorm $p_K(f) = \sup_{x \in K} |f(x)|$. Then,

$$\{p_K : K \subseteq X \text{ compact}\}$$

forms a separating family of seminorms for $C(X)$.

(b) Let G be an open subset of $\mathbb{C}$. If $H(G)$ is the set of holomorphic functions on G, then the same seminorms $\{p_K : K \subseteq G \text{ compact}\}$ forms a locally compact space whose topology is uniform convergence on compact subsets.

(c) Let X be a normed vector space. For each $\varphi \in X^*$, let $p_\varphi(x) = \left|\varphi(x)\right|$. Then, $\mathcal{P} = \left\{p_\varphi : \varphi \in X^*\right\}$ forms a separating family of seminorms for X. The topology these seminorms induce is called the *weak topology* on X. It is often denoted $\sigma(X, X^*)$.

(d) If X is a normed space, then for each $x \in X$, we may define a seminorm $p_x \colon X^* \to [0,\infty)$ by taking $p_x(\varphi) = \left|\varphi(x)\right|$. This defines a separating family of seminorms $\mathcal{P} = \{p_x : x \in X\}$, and the topology on X^* induced by these seminorms is called the weak*, or w^*, topology. It is often denoted $\sigma(X^*, X)$.

Next, we focus on an examination of the "convexity" part of the notion of locally convex topological vector spaces.

Definition: A subset $A \subseteq X$ is said to be convex if for all $v, w \in A$, the segment

$$[v,w] := \{tv + (1-t)q : 0 \leq t \leq 1\}$$

is contained in A.

Note that the intersection of any family of convex subsets is convex.

Definition: If $A \subseteq X$ is any subset, then the *convex hull* of A, denoted $\operatorname{conv}(A)$, is the intersection of all the convex subsets that contain A. If X is a TVS, the *closed convex hull* of A, denoted $\overline{\operatorname{conv}}(A)$, is the intersection of all the closed convex sets that contain A.

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References

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